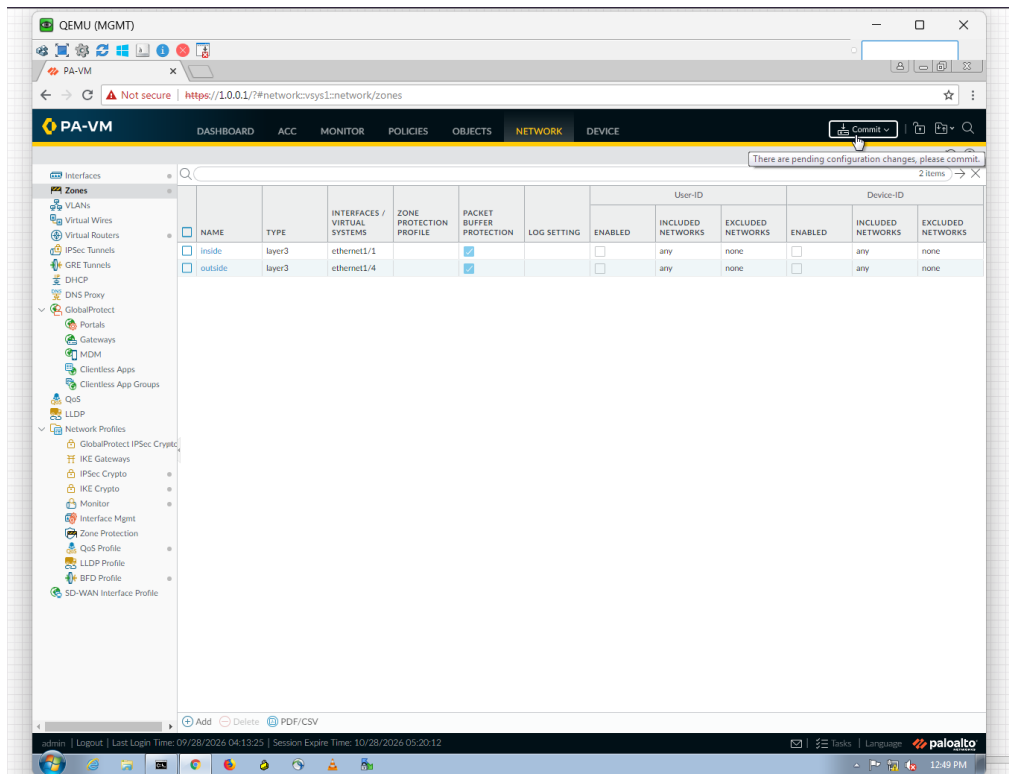


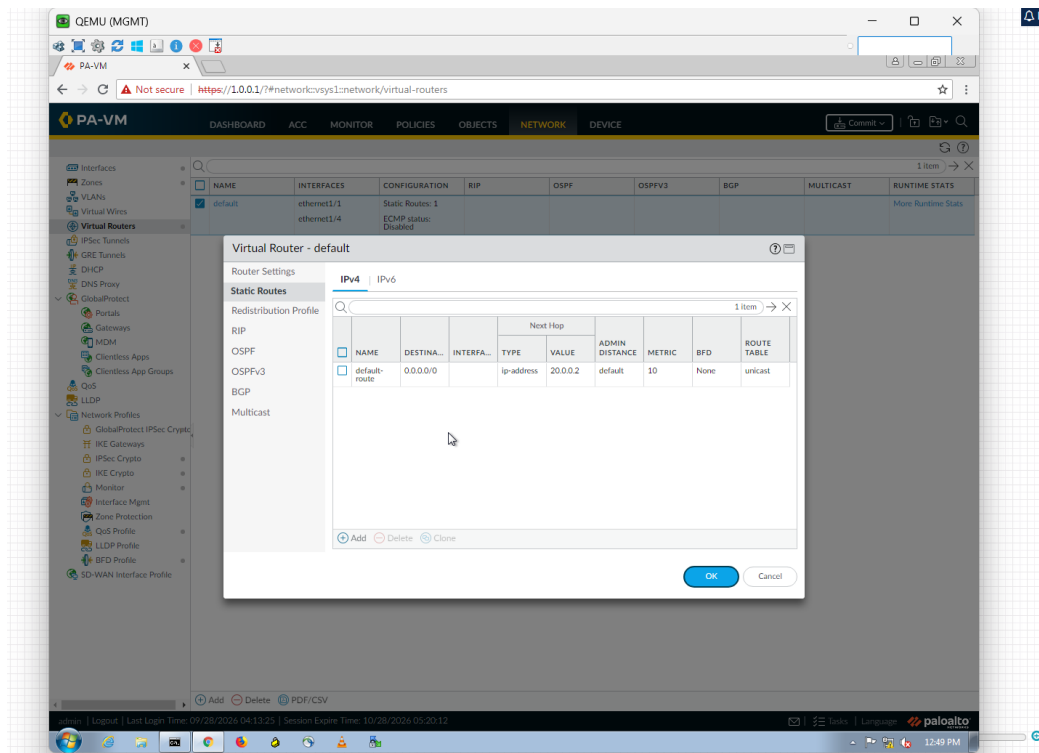
LAB 4 HIGH AVAILABILITY (HA)

GURUCHANDHRAN M

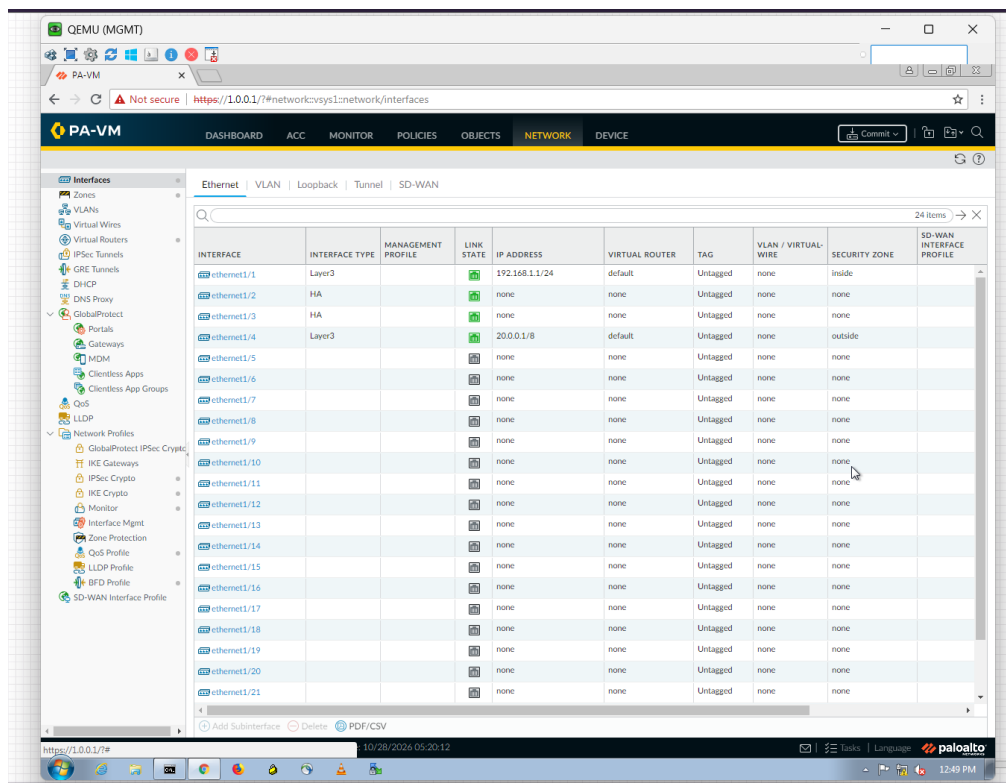
TASK 1: CONFIGURING ZONES IN THE FIREWALL



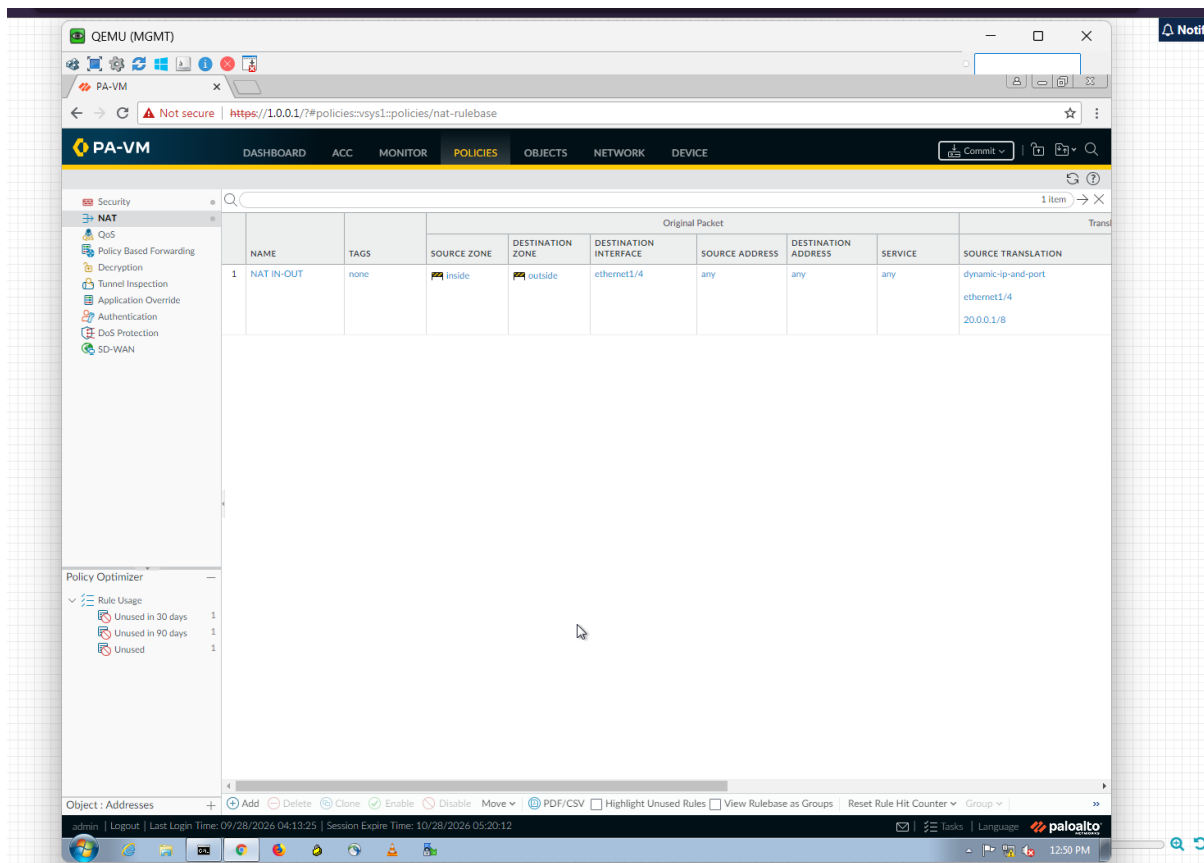
TASK 2: CONFIGURING STATIC ROUTES IN THE FIREWALL



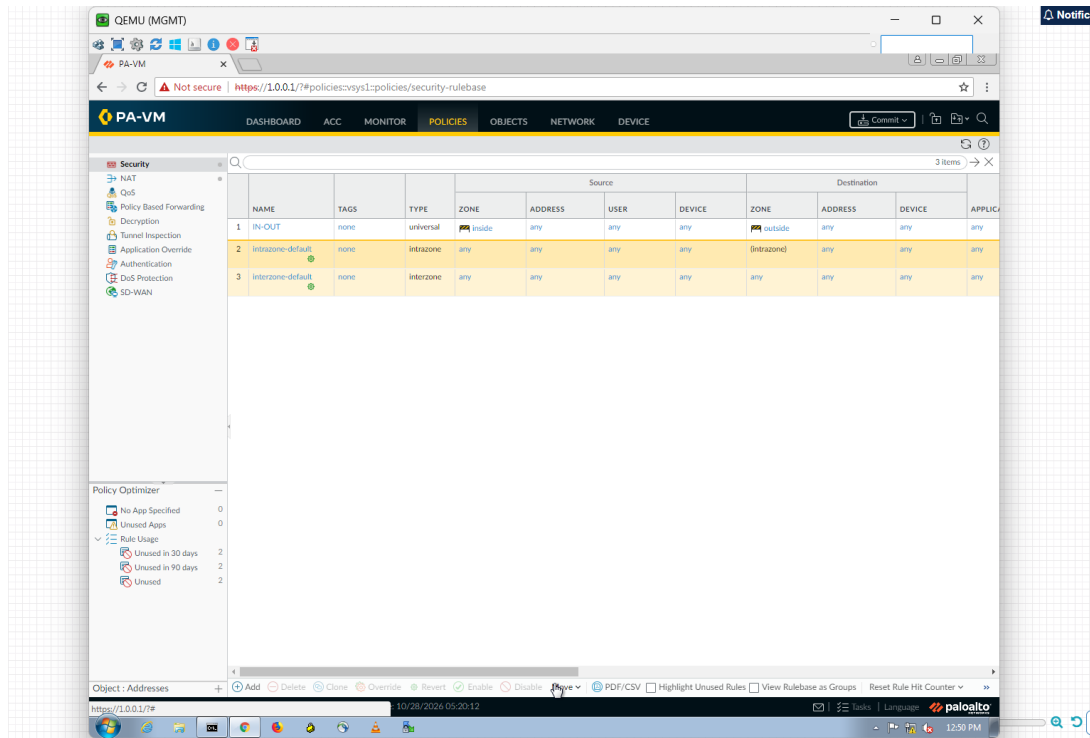
TASK 3: ASSIGNING IP ADDRESSES TO THE ETHERNET IN THE FIREWALL



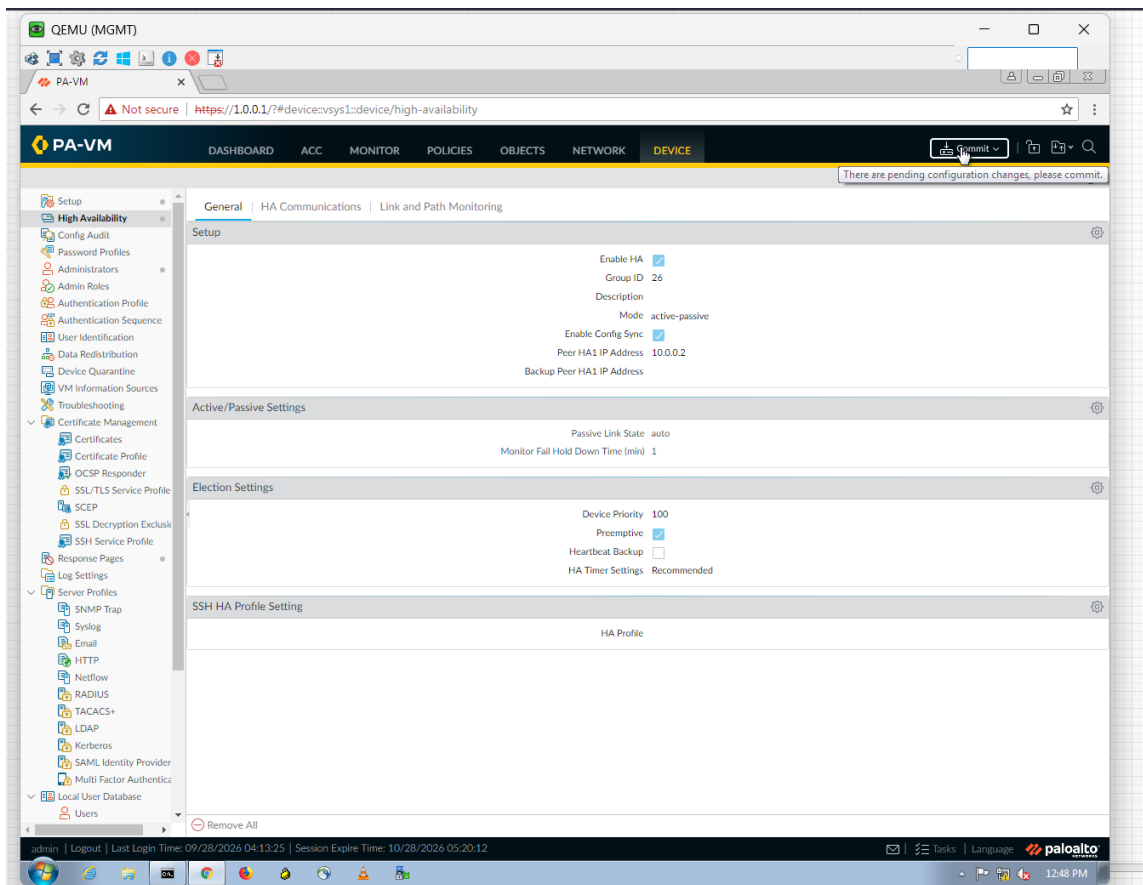
TASK 4: NAT CONFIGURATION IN THE FIREWALL



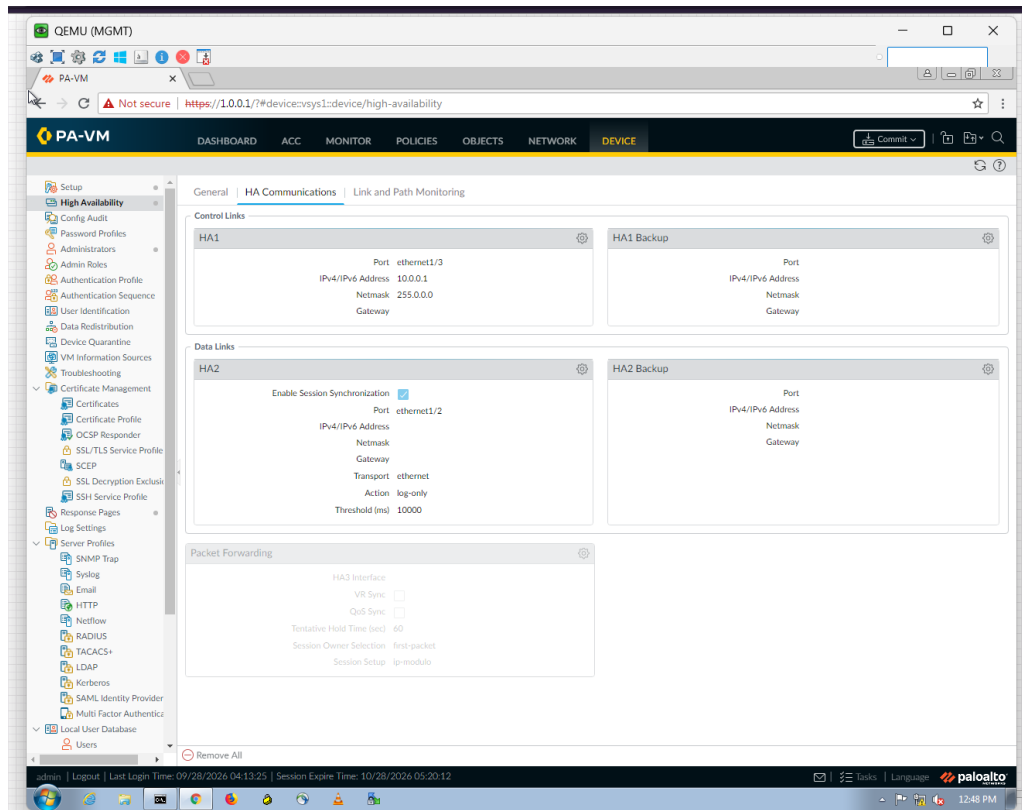
TASK 5: CONFIGURING SECURITY POLICIES IN THE FIREWALL



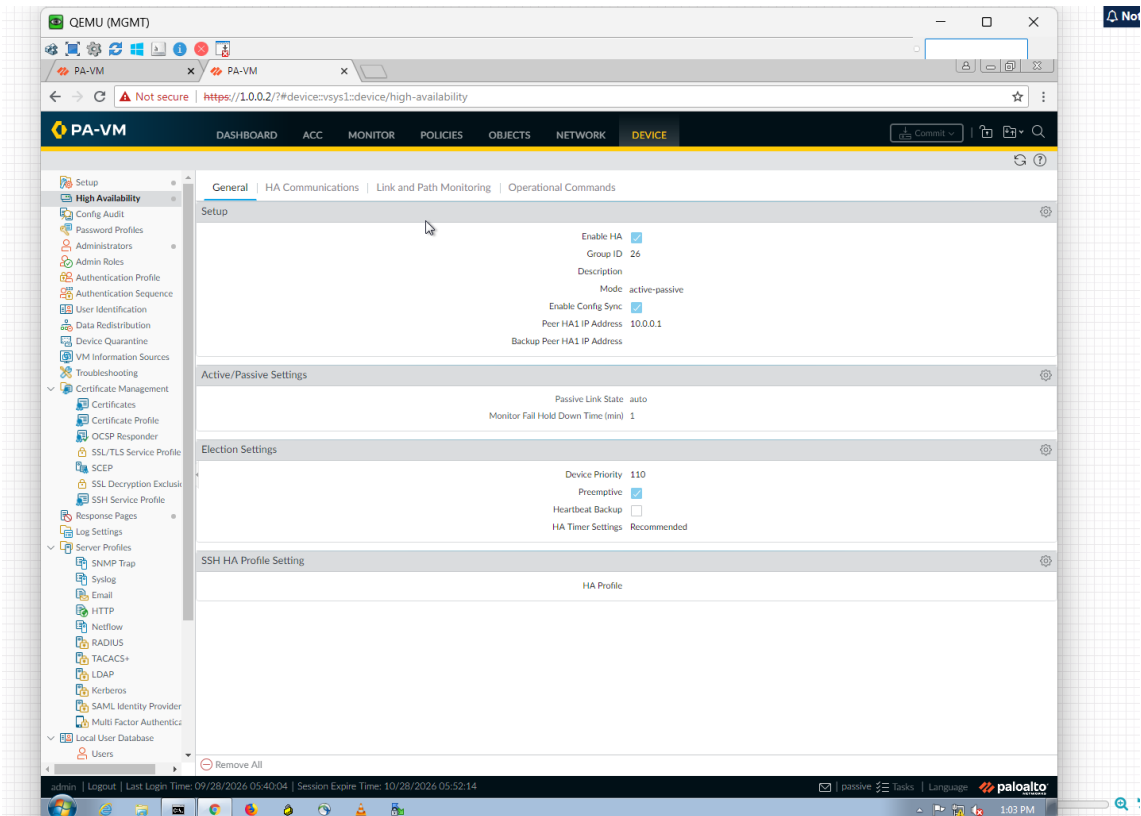
TASK 6: GENERAL HIGH AVAILABILITY CONFIGURATION IN THE FIREWALL 1



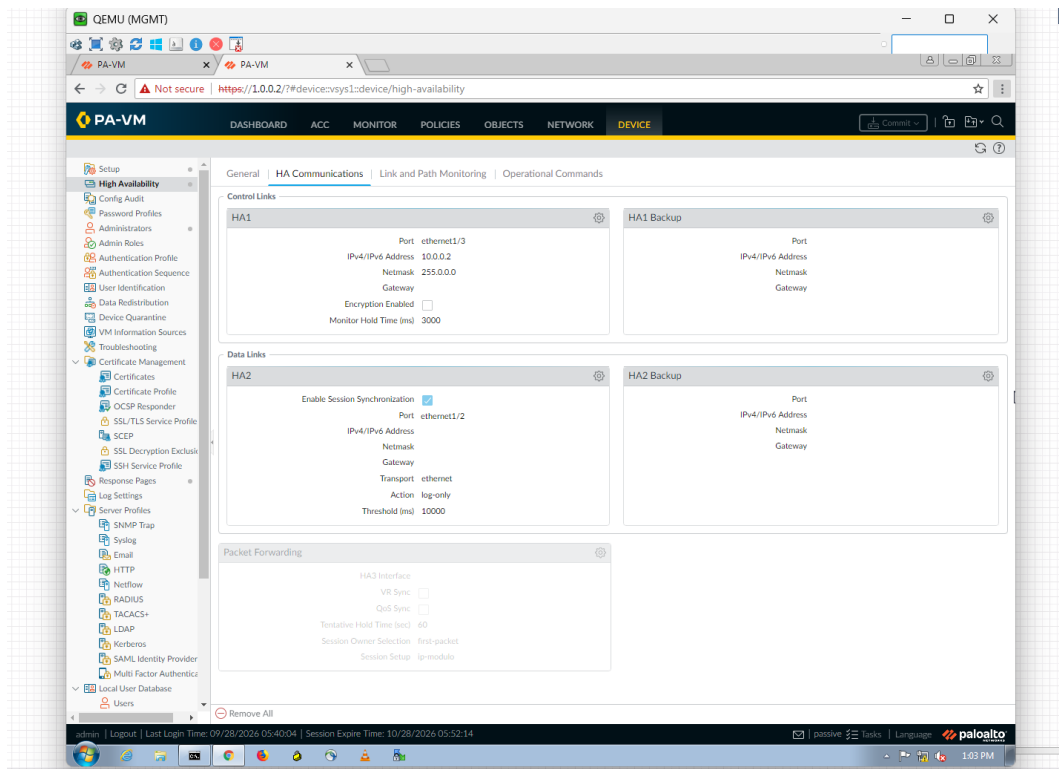
TASK 7: HA COMMUNICATION CONFIGURATION IN THE FIREWALL 1



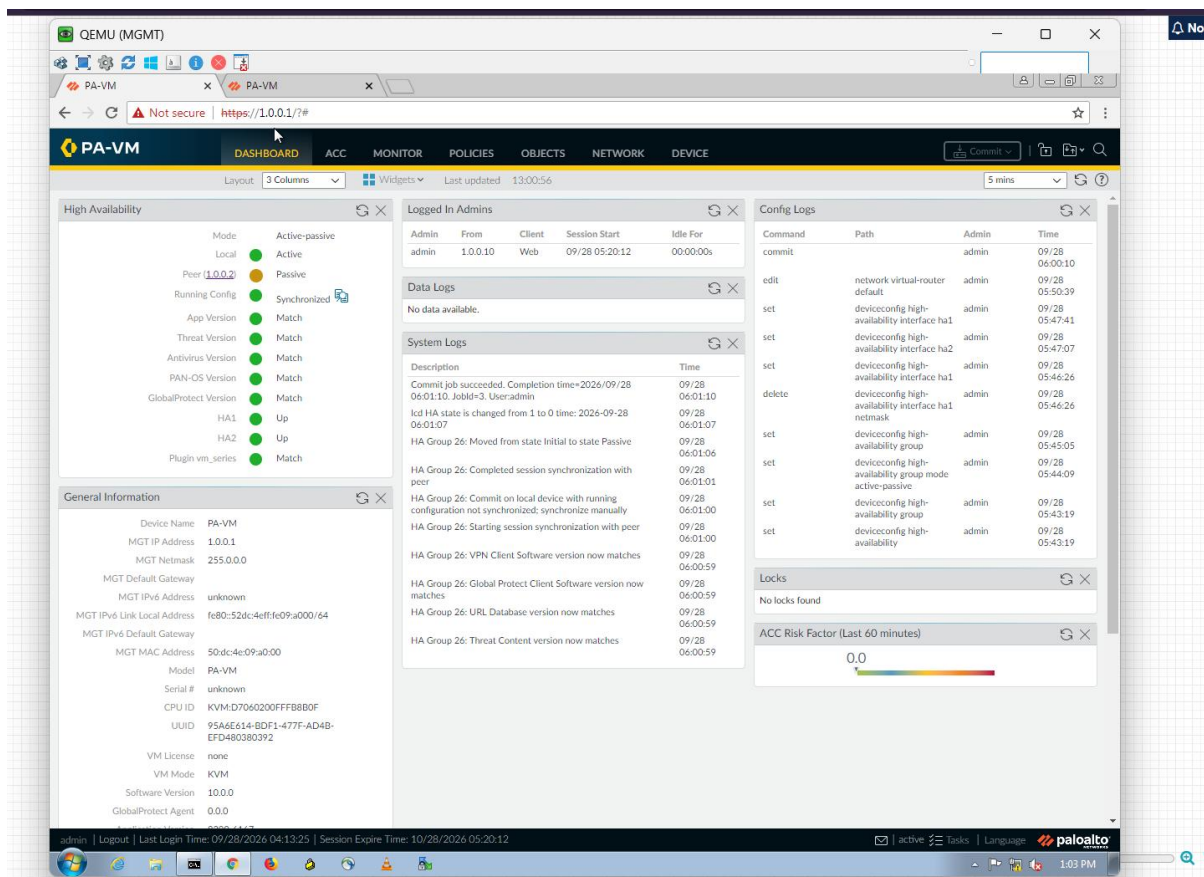
TASK 8: GENERAL HIGH AVAILABILITY CONFIGURATION IN THE FIREWALL 2



TASK 9: HA COMMUNICATION CONFIGURATION IN THE FIREWALL 2



TASK 10: VERIFYING HA IN THE FIREWALL 1 (ACTIVE)



TASK 11: VERIFYING HA IN THE FIREWALL 2 (PASSIVE)

The screenshot displays the Palo Alto VM (PA-VM) dashboard. The 'High Availability' section shows the following status:

Mode	Active-passive
Local	Passive
Peer (1.0.0.1)	Active
Running Config	Synchronized
App Version	Match
Threat Version	Match
Antivirus Version	Match
PAN-OS Version	Match
GlobalProtect Version	Match
HA1	Up
HA2	Up
Plugin vm_series	Match

The 'General Information' section provides details about the device:

Field	Value
Device Name	PA-VM
MGT IP Address	1.0.0.2
MGT Netmask	255.0.0.0
MGT Default Gateway	
MGT IPv6 Address	unknown
MGT IPv6 Link Local Address	fe80::52ad:38ff:fe0c:1200/64
MGT IPv6 Default Gateway	
MGT MAC Address	50ad:38:0c:12:00
Model	PA-VM
Serial #	unknown
CPU ID	KVM-D7060200FFB8B0F
UUID	647E6BC0-4FB8-4B29-9FFA-C0D826464789
VM License	none
VM Mode	KVM
Software Version	10.0.0
GlobalProtect Agent	0.0.0

The 'Config Logs' section shows a list of configuration changes:

Command	Path	Admin	Time
commit		admin	09/28 05:57:43
set	deviceconfig high-availability interface ha2	admin	09/28 05:56:47
set	deviceconfig high-availability interface ha1	admin	09/28 05:56:38
set	deviceconfig high-availability group	admin	09/28 05:56:11
set	deviceconfig high-availability group	admin	09/28 05:56:11
set	deviceconfig high-availability group mode active-passive	admin	09/28 05:56:02
set	deviceconfig high-availability group	admin	09/28 05:55:54
set	deviceconfig high-availability	admin	09/28 05:55:54
set	deviceconfig high-availability	admin	09/28 05:55:54
edit	network interface ethernet1/3	admin	09/28 05:55:11

TASK 12: ADDING NAT POLICIES IN THE FIREWALL 2

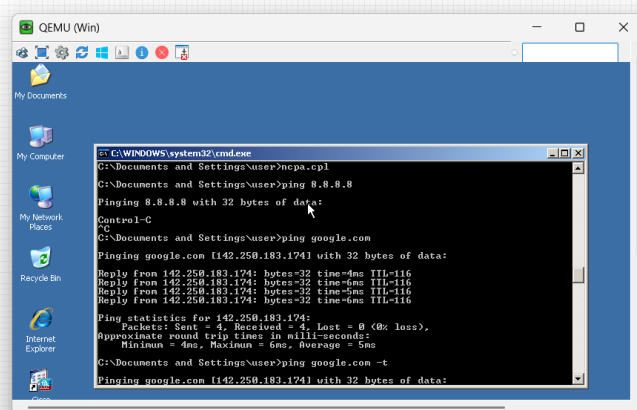
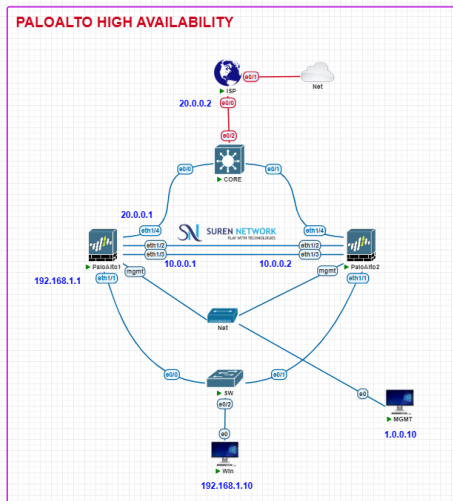
The screenshot displays the Palo Alto VM (PA-VM) dashboard, specifically the 'POLICIES' section. The 'NAT' policy is shown with the following details:

NAME	TAGS	SOURCE ZONE	DESTINATION ZONE	DESTINATION INTERFACE	SOURCE ADDRESS	DESTINATION ADDRESS	SERVICE	SOURCE TRANSLATION
1 NAT IN-OUT	none	inside	outside	ethernet1/4	any	any	any	dynamic-ip-and-port ethernet1/4 200.0.1/8

The 'Policy Optimizer' section shows the following status:

Rule Usage	Count
Unused in 30 days	1
Unused in 90 days	1
Unused	1

TASK 13: VERIFYING INTERNET REACHABILITY THROUGH FIREWALL 1



TASK 14: VERIFYING HA CONNECTIONS THROUGH ping

